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Government Visibility in Local Media Coverage of the 2025 Mount Semeru Volcanic Crisis in Indonesia

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Abstract: This study examines government visibility and the primary communicative functions of local online news during the 2025 Mount Semeru volcanic crisis in Lumajang Regency, Indonesia. Using a convergent mixed-method case-study design, it combines quantitative content analysis of 80 articles published between 19 November and 30 December 2025 across three local outlets with a focus group conducted on 21 May 2026 with eight institutional stakeholders. Actor visibility was defined by prominence in the headline, lead, and main informational purpose of each article. Government actors were foregrounded in 65 articles (81.25%). Situation/institutional updates (46.25%) and aid/logistics information (32.50%) were the most common primary functions, followed by protective-action/evacuation information (12.50%) and appeals (8.75%). Pre-reconciliation Krippendorff's alpha was 1.000 for actor, 0.885 for communication form, 0.890 for tone, and 0.675 for theme; theme findings therefore remain exploratory. Stakeholders described early information from residents and WhatsApp networks, followed by journalistic cross-checking and official verification. These accounts contextualize the production and verification of disaster news, while the content analysis provides a descriptive baseline of government prominence and message priorities within the retained corpus. The findings do not establish source dependence or communication effectiveness; warning completeness, audience comprehension, evacuation behaviour, and resilience outcomes were not measured.

Keywords: Volcanic risk communication; Local media; Government visibility; Information verification; Disaster governance; Mount Semeru

1. Introduction

Public communication is central to disaster governance. During volcanic eruptions, lahar floods, landslides, and other hazards, communities need accurate, timely, understandable, and trustworthy information alongside physical assistance. Such information guides evacuation, hazard avoidance, assistance seeking, family protection, and assessments of government action (Lindell & Perry, 2012; Wachinger et al., 2013). Disaster communication is a social, political, institutional, and cultural process connecting government agencies, media, residents, volunteers, civil society, religious leaders, and other actors in risk management (Andreastuti et al., 2023; Cottle, 2014).

Lumajang Regency's long exposure to Mount Semeru and its geographical and social characteristics make disaster communication a strategic necessity (Andreastuti et al., 2023; Ida et al., 2025). Increasing volcanic activity requires rapid information on volcano status, pyroclastic-flow runout, possible lahar floods, affected areas, evacuation routes, shelters, logistics, and assistance to affected people. Speed competes with accuracy amid inconsistent data, limited networks, and unverified social-media information (Houston et al., 2015; Veil et al., 2011). Local media are important because they are close to affected communities and familiar with the area's social and cultural context (Perreault, 2021).

Local media mediate between residents' field experiences and official statements (Cottle, 2014; Perreault, 2021), informing the public about events, responsibilities, assistance, and policies. Their proximity to local government also raises questions about watchdog performance, reproduction of government narratives, representation of victims, volunteers, community organizations and vulnerable groups, and the balance between public safety and

the success of government programmes. These questions reflect the interacting individual, routine, organizational, institutional, and broader power influences that shape news content (Entman, 1993; Reese & Shoemaker, 2016).

Disaster communication depends partly on information legitimacy. During crises, audiences seek authoritative sources, and trust in authorities and experts can influence risk perception and preparedness (Haynes et al., 2008; Wachinger et al., 2013). Government has administrative authority and official data, while media possess symbolic authority to select, highlight, organize, and disseminate information to audiences (Boholm, 2019; Entman, 1993). Heavy reliance on official statements can centre disaster discourse on government actors and narratives (Reese & Shoemaker, 2016). This may be appropriate when government is legally responsible for data, but an excessive focus on government can reduce the visibility of victims, volunteers, affected residents, and civil society (Boholm, 2019; Perreault, 2021; Tierney et al., 2006).

Recent scholarship on the Semeru eruption addresses government risk communication, disaster-risk-reduction messaging, and community engagement (Ida et al., 2025), while Indonesian volcanic-risk research emphasizes top-down warnings, bottom-up participation, and coordination among scientists, decision-makers, and communities (Andreastuti et al., 2023). The present study examines which actors become visible and which messages are prioritized in local online news, alongside stakeholders' accounts of information production and verification.

This article addresses three questions. First, how are actors, communication forms, news tone, and message content distributed in local media coverage of the Semeru eruption in Lumajang? Second, how does the prominence of government actors characterize the disaster communication discourse in the retained news corpus? Third, how does the stakeholder focus group explain the relationships among media, government, communities, and civil-society organizations in producing and verifying disaster information? These questions inform research on disaster communication, local media, and transparent, participatory, safety-oriented communication models (Andreastuti et al., 2023; Boholm, 2019).

2. Literature Review

Disaster communication covers the production, distribution, reception, and evaluation of information on risk, preparedness, emergency response, and recovery. Reynolds & Seeger (2005) emphasize fast, accurate, consistent, empathetic, action-oriented messages that reduce uncertainty, build trust, and guide protective action. In natural disasters, delayed or inconsistent communication may increase risk, generate panic, and weaken institutional legitimacy.

Comfort & Kapucu (2006) identify inter-organizational coordination as crucial to effective disaster response. Disasters are rarely handled by a single institution: local governments, disaster agencies, security forces, health services, civil society, volunteers, media, and community networks must interact within dynamic coordination systems. Communication enables these actors to develop shared understanding, align action, and reduce duplication or service gaps.

Public communication theory positions mass media within the public sphere. Habermas (1991) defines this sphere as an arena for discussing, exchanging, and contesting common issues. Local media provide space for government policy communication and a forum for citizens to monitor and criticize those policies. McQuail (2010) identifies information, correlation, continuity, entertainment, and mobilization as core media functions. Information and mobilization are especially prominent in disasters because they help communities understand danger and take appropriate action. Framing theory explains how media select aspects of reality and make them salient (Entman, 1993). Source selection, quotation placement, wording, emphasis on government action or victims' suffering, and positive, neutral, or negative tone shape public understanding. Semetko & Valkenburg (2000) identify responsibility, conflict, human interest, economic consequences, and morality frames. In disaster reporting, responsibility framing often positions government as the actor expected to explain conditions, make decisions, and provide solutions. Berelson (1952) defines content analysis as the objective, systematic, quantitative description of communication content, requiring clear categories that other researchers can replicate and apply. Holsti (1969) links it to identifying message characteristics, comparing content, and inferring production or reception conditions. Krippendorff (2013) emphasizes replicable, valid inferences from texts to their contexts, while Neuendorf (2017) stresses coding reliability and operational definitions.

Government visibility denotes a government actor's prominence in the headline, lead, and main article purpose. It does not identify quoted sources, the first source contacted, or control of reporting. Prominence may reflect formal authority, official data, news values, access constraints, or editorial choices; inferences concern representation in the retained corpus. Local media can partner with government in accelerating the distribution of accurate information while maintaining a watchdog role. Kovach & Rosenstiel (2014) identify journalism's primary task as supplying the information citizens need to govern themselves freely and responsibly. During disasters, this includes explaining risks and scrutinizing whether government responses are timely, transparent, fair, and appropriate. Dependence on government narratives may weaken scrutiny, while unverified criticism may increase panic and disinformation.

Houston et al. (2015) describe social-media uses including information seeking, warnings, assistance requests, volunteer organization, social-network restoration, and event documentation. Social media can also spread rumours and misinformation. Kent & Taylor (2002) emphasize mutuality, propinquity, empathy, risk, and commitment in dialogic communication. Disaster communication should not be one-way: agencies need feedback channels, field corrections, attention to affected groups, and recognition of other actors.

The study treats disaster news as the product of official information structures, journalistic practice, public needs, and power relations. Content analysis describes published representation; the focus group contextualizes information generation, verification, and authorization. These components are integrated interpretively and do not independently confirm the same phenomenon.

3. Methodology

3.1 Research Design and Analytical Orientation

This study uses a convergent mixed-method case-study design. Quantitative content analysis describes actor visibility, communication forms, message themes, and tone, as well as the distribution of articles by outlet and date. The stakeholder focus group interprets information exchange among residents, journalists, scientific and operational agencies, political authorities, and civil-society organizations. Frequencies describe news visibility; the focus group contextualizes information production, verification, and communication constraints.

The analysis connects media representation with volcanic risk governance by examining actor and communication-function visibility in the retained corpus and stakeholders' accounts of information production and verification. News visibility matters because repeated exposure to actors and message types can shape source recognition, the perceived legitimacy of actions, and whether preparedness and evacuation are framed as continuing risk-reduction practices or emergency responses.

3.2 Corpus and Sampling

The retained dataset contains 80 unique article-level records on the 2025 Mount Semeru eruption and related response or lahar impacts, dated 19 November-30 December 2025: 44 from Kelumajang.com, 30 from Jatimhariini.co.id, and 6 from Lumajangsatu.com. The spreadsheet preserves record number, title, date, outlet, section, primary actor, primary communication form, primary message theme, and tone. All 80 records are in the disaster section; record numbers and normalized titles are unique, and all nine analytical fields are complete. In the Supplementary Information (SI), Table A presents article metadata and Table B presents article-level coding records.

The corpus is a bounded purposive dataset, not an exhaustive or probability sample. Eligibility was defined retrospectively as: (1) a record from one of the three named local outlets; (2) publication within the bounded period; (3) substantive relevance to the Semeru eruption, lahar-related conditions, emergency response, assistance, recovery, policy, or coordination; and (4) sufficient content to assign the four primary categories. The original retrieval log lacks exact search strings, URLs, screening totals, and exclusion reasons, preventing claims of systematic capture of every eligible article. Table E in SI provides verified source links for Records 7, 40 and 60; source URLs and archived full texts have not been supplied for the remaining records. Unequal outlet totals describe the supplied corpus; cross-outlet results are descriptive and are not used to rank outlet performance.

3.3 Unit of Analysis and Coding Categories

Following Berelson's requirement for objective, systematic, quantitative description, the physical and counting unit is one complete article per spreadsheet record. The referential unit is the single actor most centrally foregrounded in the headline, lead, and main informational purpose; it measures reported actors, not source attribution. The functional unit identifies the dominant purpose: situation/institutional update, aid/logistics information, protective-action/evacuation information, or appeal. The thematic unit identifies disaster response, post-disaster recovery, disaster conditions, instructions/appeals, government policy, or inter-agency coordination. Tone captures the overall evaluation of government performance. Table C in SI provides inclusion, exclusion, and tie rules. Percentages use $N = 80$.

Categories are mutually exclusive primary classifications, not counts of every article element. Protective-action/evacuation information requires human movement away from danger, an evacuation order or status, a route, shelter, or displaced-person movement; use of 'evacuate' for object recovery is excluded. An appeal requests or recommends action but is not primarily an evacuation item. Situation/institutional updates report conditions, impacts, decisions, visits, assessments, response operations, or authorized accounts. Aid/logistics information concerns provision, distribution, need, or access to assistance. These functions do not measure message completeness, comprehension, timeliness, reach, or effectiveness.

3.4 Focus-Group Procedure and Mixed-Method Integration

Qualitative material comes from one recorded focus group held at the Lumajang branch of Nahdlatul Ulama on 21 May 2026, after the news period. Eight participants represented relevant stakeholder groups: four local journalists and one representative each from the Lumajang Office of Communication and Informatics, Nahdlatul Ulama, Muhammadiyah, and the Lumajang Indonesian Ulema Council. Discussion covered information origins, journalistic and government verification, technical and non-technical barriers, misinformation and recycled imagery, institutional transparency, media-government coordination, community networks, and proposals for an improved communication model. The group contextualizes production and verification practices; it does not estimate the prevalence of opinions among journalists, residents, or institutions.

The focus group was facilitated in Indonesian and transcribed for analysis. Participants are identified only by institutional role. Because the transcript contains automated transcription errors, the analysis uses short passages whose meaning is clear from the surrounding context, with translations prepared by the authors. Thematic grouping used deductive codes for information origin, verification, authority, barriers, coordination, transparency, participation, and channel redundancy. Passages were checked against the supplied transcript; the audio recording was unavailable for this revision. The material reflects stakeholder interpretations and does not establish audience effects or validate the quantitative coding.

3.5 Data Quality, Reliability, and Ethical Boundaries

Data checks addressed date consistency, uniqueness, completeness, and frequency totals across the 80 retained records. The documented coding corrections include reassigning record 40 from other government institution to volunteer/non-governmental organization (NGO) because Ansor and Banser are civil-society organizations. Record 7 was reassigned from appeal to protective-action/evacuation information because it explicitly urged red-zone residents to evacuate. Record 60 was reassigned from protective-action/evacuation information to situation/institutional update because it concerns the recovery of a mosque drum rather than human evacuation. Section B in SI provides the article-level coding records underlying the reported frequencies.

Intercoder reliability was assessed using a simple random subsample of 16 articles, representing 20% of the corpus (record IDs 1, 5, 11, 13, 19, 23, 29, 31, 32, 34, 36, 46, 50, 68, 69, and 71). Table 1 reports agreement and Krippendorff's alpha for each variable.

Table 1. Intercoder reliability for the 20% subsample (n = 16)

Variable	Agreement	Observed Disagreement	Expected Disagreement	Krippendorff's alpha
Primary actor	16/16 (100.0%)	0.0000	0.5565	1.000
Communication form	15/16 (93.8%)	0.0625	0.5423	0.885
Tone	15/16 (93.8%)	0.0625	0.5665	0.890
Message theme	12/16 (75.0%)	0.2500	0.7681	0.675

Note: n = reliability subsample size.

A second trained coder independently coded the same complete articles using the operational codebook and dominant-category rule, without seeing the first coder's decisions. Krippendorff's alpha for nominal data was calculated separately for primary actor, communication form, message theme, and tone before discussion or reconciliation of coding disagreements, using all 32 coder assignments per variable. Observed disagreement was compared with that expected from both coders' pooled marginal distributions. Alpha values of at least 0.80 were treated as satisfactory; values from 0.67 to 0.79 as marginal and suitable only for tentative interpretation (Krippendorff, 2013). Table D in SI reproduces the complete two-coder decision matrix.

These coefficients describe the earlier coding exercise and do not by themselves establish category validity. Because theme reliability was marginal and no new independent round was available, theme frequencies remain exploratory descriptors; detailed theme-based inference is avoided.

4. Results

4.1 Corpus Distribution and Government Prominence

The retained corpus contains 80 article records: 44 from Kelumajang.com (55.0%), 30 from Jatimhariini.co.id (37.5%), and 6 from Lumajangsatu.com (7.5%). Table 2 presents the distribution of these publications across six seven-day windows: 24, 21, 14, 17, 3, and 1 records. These counts describe the retained corpus, not the hazard trajectory, because they were not linked to daily alert levels or eruption events.

Table 2. Temporal distribution of article records (*N* = 80)

Seven-Day Window	Kelumajang	Lumajangsatu	Jatimhariini	Total
19–25 Nov	17	3	4	24
26 Nov–2 Dec	13	1	7	21
3–9 Dec	3	1	10	14
10–16 Dec	9	1	7	17
17–23 Dec	1	0	2	3
24–30 Dec	1	0	0	1
Total	44	6	30	80

Note: *N* = total sample size.

Government actors were foregrounded in 65 articles (81.25%) and non-government actors in 15 (18.75%) (Table 3). These counts measure prominence in the retained news agenda, not source dependence or control over reporting.

Table 3. Primary actor by outlet (*N* = 80)

Actor	Kelumajang	Lumajangsatu	Jatimhariini	Total	Percentage (%)
Regent of Lumajang	7	5	8	20	25.00
BPBD	1	0	1	2	2.50
Other government institutions	30	1	12	43	53.75
Volunteers/NGOs	4	0	2	6	7.50
Community members	2	0	7	9	11.25
Total	44	6	30	80	100.00

Note: *N* = total sample size. BPBD = Badan Penanggulangan Bencana Daerah (Regional Disaster Management Agency), referring here to the Lumajang agency. NGOs = non-governmental organizations.

The aggregate category ‘other government institutions’ includes technical, political, health, security, and administrative bodies. It cannot support comparisons among these types or establish that a named agency was absent from article text. The two primary-actor assignments for Lumajang’s Regional Disaster Management Agency (Badan Penanggulangan Bencana Daerah; BPBD) do not measure its total quotation frequency or operational involvement.

4.2 Tone and Accountability

Positive and neutral coverage each accounted for 34 articles (42.5%), while 12 (15.0%) were negative. Table 4 presents the outlet distributions. Because sample sizes differ substantially, the counts should not be used to rank editorial independence.

Table 4. News tone by outlet (*N* = 80)

Tone	Kelumajang	Lumajangsatu	Jatimhariini	Total	Percentage (%)
Positive	22	3	9	34	42.5
Neutral	16	3	15	34	42.5
Negative	6	0	6	12	15.0
Total	44	6	30	80	100.0

Note: *N* = total sample size.

Overall, 85% of coverage was positive or neutral; tone coding showed satisfactory intercoder reliability (alpha = 0.890). Non-sensational reporting may support public confidence, while limited negative coverage may narrow scrutiny of warning failures, unsafe access, unequal assistance, relocation disputes, and conflicting information. Focus-group accounts reflected this tension: official information supported verification, but participants also described uneven detail, transparency, and representation of field perspectives.

4.3 Communication Forms and Actionable Safety Information

Table 5 reports 37 situation/institutional updates (46.25%), 26 aid/logistics articles (32.5%), 10 protective-action/evacuation articles (12.5%), and 7 appeals (8.75%). Updates and appeals occurred across all three outlets; aid/logistics and protective-action/evacuation items appeared in Kelumajang and Jatimhariini.

These frequencies describe primary communication functions. They do not establish the completeness, timeliness, reach, or comprehension of safety information. Articles assigned to other categories may contain secondary safety details that the mutually exclusive coding does not capture.

The outlet distribution is an agenda measure under the primary-function rule, not a ranking of safety performance. Assessing warning completeness would require separate coding of hazard, location, timing, action, route, shelter, assistance, and timestamps.

Table 5. Primary communication form by outlet (*N* = 80)

Communication Form	Kelumajang	Lumajangsatu	Jatimhariini	Total	Percentage (%)
Situation/institutional update	19	4	14	37	46.25
Aid/logistics information	19	0	7	26	32.5
Protective-action/evacuation information	2	0	8	10	12.5
Appeal	4	2	1	7	8.75
Total	44	6	30	80	100.0

Note: *N* = total sample size.

4.4 Message Themes Across the Disaster Cycle

Disaster response was the most frequent theme (39; 48.75%) and predominated in all three outlets (Table 6). No separate primary category captured preparedness information such as hazard zones, evacuation routes, drills, household or livelihood plans, accessible transport, or lahar triggers. With marginal theme reliability (alpha = 0.675), distinctions among response, policy, recovery, and related themes remain exploratory. These distributions describe the coded sample and do not establish causal relationships or coverage outside the sampled period.

Table 6. Primary message theme by outlet (*N* = 80)

Message Theme	Kelumajang	Lumajangsatu	Jatimhariini	Total	Percentage (%)
Disaster response	21	2	16	39	48.75
Post-disaster recovery	7	1	5	13	16.25
Disaster conditions	3	1	7	11	13.75
Instructions/appeals	7	2	1	10	12.5
Government policy	5	0	0	5	6.25
Inter-agency coordination	1	0	1	2	2.5
Total	44	6	30	80	100.0

Note: *N* = total sample size.

4.5 Information Flow: Community Signals, Journalistic Verification, and Official Authority

Table 7 places the quantitative patterns alongside complementary stakeholder accounts, which provide context rather than direct evidence of source use or the causes of the observed frequencies.

Table 7. Quantitative patterns and complementary stakeholder accounts

Quantitative Pattern	Stakeholder Account	Interpretation and Limits
Government actor prominence: 81.25%	"One door" for official casualty and impact figures (government communication representative).	Official confirmation was valued; prominence does not establish source dependence.
Protective-action/evacuation primary function: 12.5%	Netizen reports checked with BPBD or police to avoid panic (Journalist 3).	Verification was emphasized; warning completeness and audience effects were not measured.
Situation/institutional updates: 46.25%	Tension between speed and accuracy (government communication representative).	This tension was perceived, not quantitatively assessed.
Aid/logistics information: 32.5%	Unclear or overlapping data can lead to unequal aid distribution (Muhammadiyah disaster-management participant).	A coordination concern was reported; its prevalence was not established.

Note: Percentages reflect primary classifications (total sample size of 80). Stakeholder accounts summarize the focus-group transcript. BPBD = Badan Penanggulangan Bencana Daerah (Regional Disaster Management Agency), referring here to the Lumajang agency.

A local journalist stated, 'We do not immediately publish information from netizens; we confirm it with BPBD or the police so it does not create panic' (Journalist 3, author translation). Another described residents and WhatsApp groups as early sources, followed by verification with BPBD and affected residents (Journalist 2, author translation). These accounts suggest a verification sequence without establishing its frequency across the 80 articles. A government communication representative emphasized a 'one-door' protocol for official casualty and impact figures.

Participants reported electricity and mobile-signal disruption, hazardous access, and recycled social-media material. A civil-society participant noted that older events could be presented as current news. The government communication representative described ‘fast but inaccurate versus accurate but late’, while a Muhammadiyah disaster-management participant linked unclear and overlapping data to unequal aid distribution. These accounts explain the participants’ emphasis on verification, timestamps, and fallback channels.

5. Discussion

5.1 Government-Centred Communication as Both Resource and Risk

Government actors were foregrounded in 81.25% of articles, a pattern consistent with their formal authority and operational responsibilities. This actor measure does not identify initial informants, count quoted sources, or establish journalists’ dependence on government. Situation/institutional updates and logistics were more often primary than protective-action/evacuation information or appeals. This measures agenda prominence, not safety-message completeness or guidance available through other articles, broadcasts, village channels, and direct warnings. Low primary-actor counts for BPBD or other agencies do not establish institutional absence. Future studies should code all quoted sources and distinguish technical, political, health, security, and administrative institutions before assessing the distribution of communicative authority.

5.2 Evacuation Decision Support and Public Safety

Evacuation messages must specify the hazard, affected locations, triggers, departure time, safe routes and shelters, update time, and verification contact. They must also specify assistance arrangements for children, older people, persons with disabilities, tourists, livestock, and residents without transport. Ten records (12.5%) primarily concerned protective action or evacuation, and seven (8.75%) were appeals. These frequencies measure agenda visibility, not warning quality, compliance, or behavioural effects. Evaluation requires element-level coding and evidence of exposure, comprehension, and action. Focus-group accounts suggest that local media can bridge field reporting and official verification. Evacuation protocols should give journalists rapid access to timestamped hazard information, reproducible maps, named spokespersons, and correction channels, while providing authorities with verified reports of blocked routes, isolated settlements, outages, and households unable to evacuate. The effects of this two-way system require evaluation.

5.3 Preparedness and the Temporal Imbalance of Coverage

Exploratory coding assigned 48.75% of articles to disaster response and 2.5% to coordination. With theme alpha of 0.675 and a response-period corpus, these descriptive values cannot establish longer-term preparedness performance. Preparedness also builds trust: residents need to understand technical agencies, alert levels, hazard zones, and protective measures before urgent warnings arrive. Local media can explain scientific information and uncertainty, distinguish eruption hazards from rain-triggered lahars, and clarify why secondary hazards and access restrictions may persist as volcanic activity declines.

5.4 Community Knowledge, Inclusion, and Resilience

Community members and volunteers/NGOs were foregrounded in 18.75% of articles, although focus-group accounts described residents and local networks as early information sources. This contrast raises questions about how narrative visibility reflects informational contributions. Recognising local knowledge, reciprocal trust, and affected groups’ influence on relief and recovery priorities may support resilience. Inclusion requires structured authentication of community observations. Village focal points, trained volunteers, religious and social organisations, and local journalists can provide geolocated reports, photographs, access conditions, and priority needs. Technical agencies remain responsible for validation and hazard interpretation; this process should preserve source attribution and residents’ participation.

Relocation resistance may reflect concerns about land, livelihoods, livestock, social networks, schooling, or long-term support. Treating it solely as non-compliance may overlook these barriers. Dialogue can clarify why protective advice is difficult to follow and inform transport, livelihood, housing, and social-support measures that make safer choices feasible.

5.5 Toward an Integrated Volcanic Risk Communication Architecture

Stakeholder accounts and content patterns motivate a proposed architecture linking scientific, emergency-management, communication, village, media, and civil-society actors. Its feasibility and effects require testing

before it can be considered an evidence-based intervention.

Evacuation messages should follow the requirements outlined in Section 5.2, with media-ready maps and brief plain-language versions accompanying technical bulletins. Redundant channels are needed when electricity or networks fail. Local radio, village loudspeakers, mosque networks, mobile public-address units, community volunteers, and printed route markers can complement WhatsApp and online media to maintain last-mile communication.

Community feedback should be institutionalised through verified reports of blocked routes, isolated households, unmet logistical needs, misinformation, and changing conditions. Reports should reach operational command structures rather than remain in informal messaging groups.

Preparedness evaluation should assess whether residents recognise official sources, interpret alert levels, locate evacuation routes, and act under realistic scenarios. Content frequencies describe agendas; assessing effectiveness requires audience-centred and behavioural evidence.

5.6 Contribution to Natural-Hazard Research

The study provides a descriptive baseline of actor prominence and communicative functions in local volcanic-crisis reporting. Stakeholder accounts contextualize reporting practices, while the content measures remain distinct from source attribution.

6. Theoretical and Practical Implications

The findings highlight the need to distinguish source authority from communicative adequacy. Official credibility alone does not ensure actionable, inclusive, timely, or accessible information during infrastructure disruption. Assessing government prominence therefore requires evidence of its communication consequences, rather than assumptions about coordination or political domination.

Practitioners should repeat actionable safety content alongside verified institutional information. Updates should help people decide whether to evacuate, avoid river channels, use respiratory protection, postpone travel, access shelter, request assistance, or prepare for secondary lahars. Technical agencies should be named as sources, with political leaders reinforcing their guidance.

Media, journalist associations, and government communication agencies can develop reporting protocols covering source hierarchy, timestamps, corrections, maps, victim privacy, journalist safety, and community inclusion. Briefings should extend beyond acute emergencies. Exercises should assess whether warnings can be delivered consistently through radio, village, religious, and volunteer channels when networks fail.

7. Limitations and Future Research

Seven limitations constrain interpretation: the purposive corpus covers three outlets with unequal totals; most URLs and archived texts are missing, as are search strings, screening counts, and exclusion logs; actor coding does not separately identify quoted sources; each variable retains one primary category; warning completeness was not coded; theme reliability was marginal (alpha = 0.675), with no further independent assessment of revised classifications; and the post-period focus group included institutional stakeholders rather than affected residents, with quotations unchecked against unavailable audio. These limitations preclude causal or effectiveness claims.

Future studies should retain retrieval and screening logs, archive article URLs or texts, code warning completeness, and refine thematic rules before an independent reliability round. Rules should distinguish immediate operations, administrative decisions, and longer-term recovery by time horizon and object of action. Studies should include affected residents, align official warnings with media publication, examine non-digital channels, and test relationships among exposure, trust, preparedness, and evacuation.

8. Conclusion

The 80-article corpus documents government prominence (81.25%) during the 2025 Mount Semeru crisis. Situation/institutional updates (46.25%) and aid/logistics (32.5%) were the most common primary functions; protective-action/evacuation information (12.5%) and appeals (8.75%) were less often primary.

Stakeholders described community and WhatsApp signals, journalistic verification, official confirmation, misinformation, and outages. Their accounts contextualize news patterns but do not establish system effectiveness or uniform practices across the corpus.

Clearer protective-action content, source-specific coding, redundant channels, and structured community feedback warrant evaluation; the study does not demonstrate that this architecture improves emergency response or resilience.

Author Contributions

Conceptualization, B.P. and R.F.R.; methodology, B.P. and R.F.R.; software, B.P. and M.M.; validation, B.P., R.F.R., M.M., and H.A.J.; formal analysis, B.P., R.F.R., and M.M.; investigation, B.P., R.F.R., M.M., and H.A.J.; resources, B.P.; data curation, B.P. and M.M.; writing—original draft preparation, B.P. and R.F.R.; writing—review and editing, B.P., R.F.R., M.M., and H.A.J.; visualization, B.P. and H.A.J.; supervision, B.P.; project administration, B.P.; funding acquisition, B.P. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement

Audio-recorded verbal informed consent was obtained from all eight adult professional participants for participation in the study and publication of de-identified quotations or paraphrases.

Ethical Approval

Institutional authorization approval were provided through Notice No. B/4964/UN38.16/TU.00.02/2026, signed by the Head of LPPM, Universitas Negeri Surabaya, on 15 April 2026. This authorization did not constitute review by a separately constituted research ethics committee.

Data Availability

The article-level corpus metadata, coding records, operational codebook, and paired coder decisions are included in the Supplementary Information. Additional retained research materials may be shared upon reasonable request, subject to participant confidentiality and institutional data-management requirements.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors used ChatGPT (OpenAI) for English-language editing, manuscript organization, and computational checking of the intercoder-reliability matrix. It was not used to generate coder decisions, collect source articles, or conduct the focus group. The authors reviewed and verified all AI-assisted revisions, calculations, factual claims, and references and accept full responsibility for the accuracy, originality, and integrity of the final manuscript.

References

Andreastuti, S. D., Paripurno, E. T., Subandriyo, S., Syahbana, D. K., & Prayoga, A. S. (2023). Volcano disaster risk management during crisis: Implementation of risk communication in Indonesia. *J. Appl. Volcanol., 12*, 3. https://doi.org/10.1186/s13617-023-00129-2.

Berelson, B. (1952). *Content Analysis in Communication Research*. Free Press.

Boholm, Å. (2019). Risk communication as government agency organizational practice. *Risk Anal., 39*(8), 1695–1707. https://doi.org/10.1111/risa.13302.

Comfort, L. K. & Kapucu, N. (2006). Inter-organizational coordination in extreme events: The World Trade Center attacks, September 11, 2001. *Nat. Hazards, 39*(2), 309–327. https://doi.org/10.1007/s11069-006-0030-x.

Cottle, S. (2014). Rethinking media and disasters in a global age: What's changed and why it matters. *Media War Confl., 7*(1), 3–22. https://doi.org/10.1177/1750635213513229.

Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *J. Commun.*, *43*(4), 51–58. https://doi.org/10.1111/j.1460-2466.1993.tb01304.x.
Habermas, J. (1991). *The Structural Transformation of the Public Sphere: An Inquiry Into a Category of Bourgeois Society*. MIT Press.
Haynes, K., Barclay, J., & Pidgeon, N. (2008). The issue of trust and its influence on risk communication during a volcanic crisis. *Bull. Volcanol.*, *70*(5), 605–621. https://doi.org/10.1007/s00445-007-0156-z.
Holsti, O. R. (1969). *Content Analysis for the Social Sciences and Humanities*. Addison-Wesley.
Houston, J. B., Hawthorne, J., Perreault, M. F., Park, E. H., Goldstein Hode, M., Halliwell, M. R., Turner McGowen, S. E., Davis, R., Vaid, S., McElderry, J. A., et al. (2015). Social media and disasters: A functional framework for social media use in disaster planning, response, and research. *Disasters*, *39*(1), 1–22. https://doi.org/10.1111/disa.12092.
Ida, R., Gunawan, E., Widiyantoro, S., Pratama, C., Hanifa, N. R., & Saud, M. (2025). Disaster risk reduction communication during the Mount Semeru eruption in East Java, Indonesia. *Jàmbá*, *17*(1), a1849. https://doi.org/10.4102/jamba.v17i1.1849.
Kent, M. L. & Taylor, M. (2002). Toward a dialogic theory of public relations. *Public Relat. Rev.*, *28*(1), 21–37. https://doi.org/10.1016/S0363-8111(02)00108-X.
Kovach, B. & Rosenstiel, T. (2014). *The Elements of Journalism: What Newspeople Should Know and the Public Should Expect* (3rd ed.). Three Rivers Press.
Krippendorff, K. (2013). *Content Analysis: An Introduction to Its Methodology* (3rd ed.). SAGE Publications.
Lindell, M. K. & Perry, R. W. (2012). The protective action decision model: Theoretical modifications and additional evidence. *Risk Anal.*, *32*(4), 616–632. https://doi.org/10.1111/j.1539-6924.2011.01647.x.
McQuail, D. (2010). *McQuail's Mass Communication Theory* (6th ed.). SAGE Publications.
Neuendorf, K. A. (2017). *The Content Analysis Guidebook (2nd ed.).* SAGE Publications. https://doi.org/10.4135/9781071802878.
Perreault, M. F. (2021). Journalism beyond the command post: Local journalists as strategic citizen stakeholders in natural disaster recovery. *Journal. Stud.*, *22*(10), 1279–1297. https://doi.org/10.1080/1461670X.2021.1950565.
Reese, S. D. & Shoemaker, P. J. (2016). A media sociology for the networked public sphere: The hierarchy of influences model. *Mass Commun. Soc.*, *19*(4), 389–410. https://doi.org/10.1080/15205436.2016.1174268.
Reynolds, B. & Seeger, M. W. (2005). Crisis and emergency risk communication as an integrative model. *J. Health Commun.*, *10*(1), 43–55. https://doi.org/10.1080/10810730590904571.
Semetko, H. A. & Valkenburg, P. M. (2000). Framing European politics: A content analysis of press and television news. *J. Commun.*, *50*(2), 93–109. https://doi.org/10.1111/j.1460-2466.2000.tb02843.x.
Tierney, K., Bevc, C., & Kuligowski, E. (2006). Metaphors matter: Disaster myths, media frames, and their consequences in Hurricane Katrina. *Ann. Am. Acad. Pol. Soc. Sci.*, *604*(1), 57–81. https://doi.org/10.1177/0002716205285589.
Veil, S. R., Buehner, T., & Palenchar, M. J. (2011). A work-in-process literature review: Incorporating social media in risk and crisis communication. *J. Conting. Crisis Manag.*, *19*(2), 110–122. https://doi.org/10.1111/j.1468-5973.2011.00639.x.
Wachinger, G., Renn, O., Begg, C., & Kuhlicke, C. (2013). The risk perception paradox—Implications for governance and communication of natural hazards. *Risk Anal.*, *33*(6), 1049–1065. https://doi.org/10.1111/j.1539-6924.2012.01942.x.

Supplementary Information

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